AI Systems Engineering – a new engineering discipline?

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Personal Profile

Dr.-Ing. Thomas Usländer, Fraunhofer IOSB

- Diploma in Computer Science, University of Karlsruhe (1981), PhD at KIT (2010)
- Department „Information Management and Production Control (ILT)“
- Business Developer AI Systems Engineering
- Project Manager AI Alliance Baden-Württemberg eG
- Project Manager CC-KING – Competence Center Karlsruhe for AI Systems Engineering
- Co-chair of the IDS-Industrial community
- Active member of various Industrie 4.0 committees and projects

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Digital Production (AI) Ecosystems

PLATTFORM INDUSTRIE 4.0

open and secure system architectures
making internationally accepted standards usable

INTERNATIONAL DATA SPACES ASSOCIATION

Data spaces

AI Systems

Physical World

Virtual World

interoperability – physical assets

interoperability – virtual assets

gaia-x

MANUFACTURING-X

Catena-X Automotive Network

FACTORY-X

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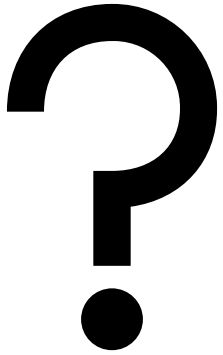
Motivation

AI only becomes an innovation when it is applied.

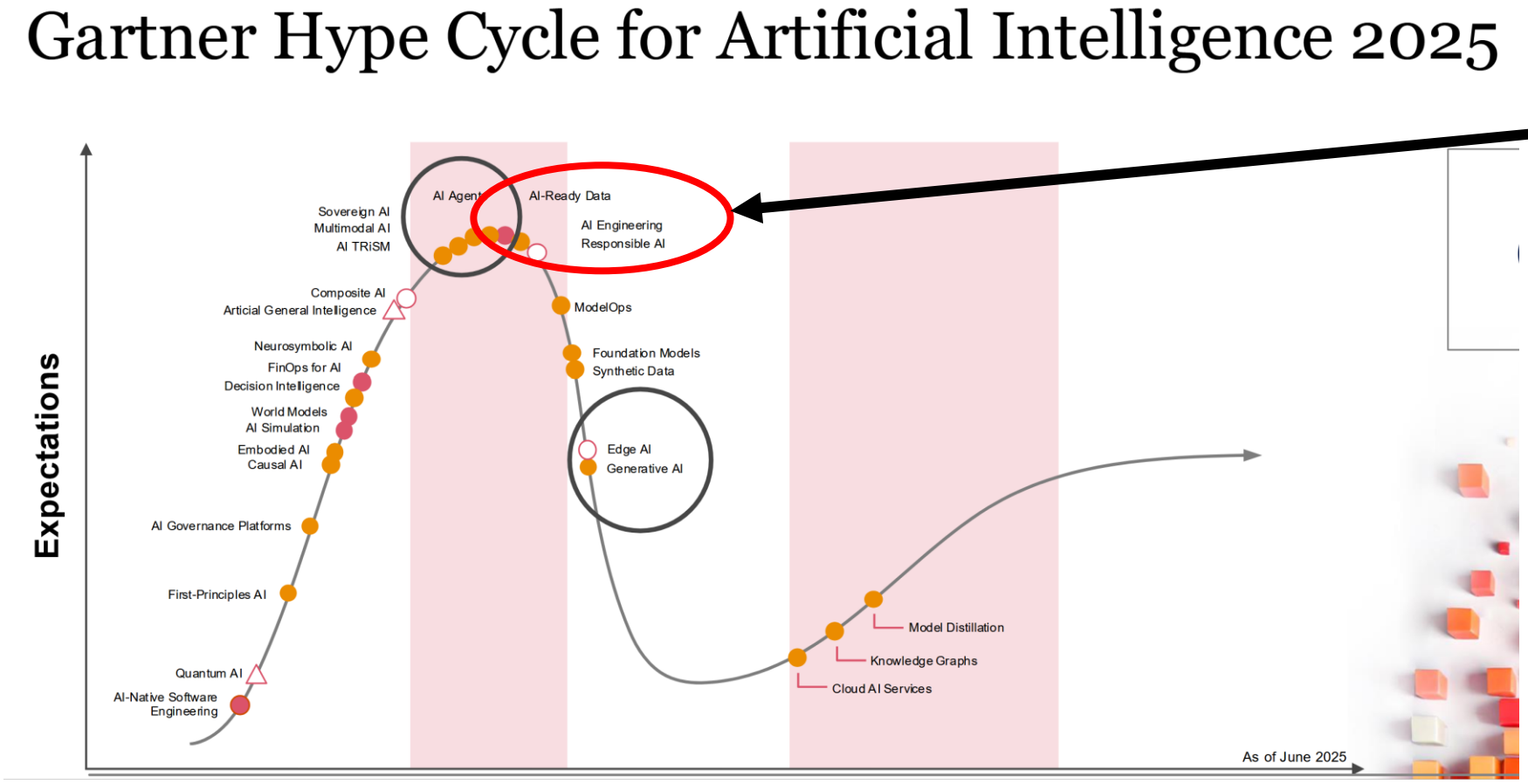


reliably
securely
robustly
efficiently
…

Motivation global

Gartner (2024): only 9% of organizations are currently AI-mature

MIT (2025): 95% of the genAI projects fail



Demand today:

- AI-ready data
- AI (Systems) Engineering + responsible AI

Commodity in 5 years

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Gartner Survey (2024): https://www.gartner.com/en/newsroom/press-releases/2024-05-07-gartner-survey-finds-generative-ai-is-now-the-most-frequently-deployed-ai-solution-in-organizations

Differentiating factors: Focus on four foundational capabilities:

- A scalable AI operating model, balancing centralized and distributed capabilities.
- A focus on AI engineering, designing a systematic way of building and deploying AI projects into production.
- An investment on upskilling and change management across the wider organization.
- A focus on trust, risk and security management (TRiSM) capabilities to mitigate the risks that come from AI implementations and drive better business outcomes.

Gartner Glossary

AI engineering is foundational for enterprise delivery of AI solutions at scale.

The discipline unifies DataOps, MLOps and DevOps pipelines to create coherent enterprise development, delivery (hybrid, multicloud, edge), and operational (streaming, batch) AI-based systems.

Data space projects: Catena-X and Factory-X

Catena-X
Automotive Network

- Lighthouse project for the automotive industry
  → Data sharing within the entire ecosystem
- Builds data driven value chains
- Makes business applications interoperable
- Focusses on time to value
- Guides and supports user groups

https://youtu.be/oCqAYa_E8zY

FACTORY-X

- Lighthouse project for the ‚factory outfitters': machine builders, component suppliers, automation providers and software partners
- Data sharing among factories and partners of the ecosystem
- Use cases along the supply chain and for run time data gathered during operation
- → Improve competitiveness, resiliency and product carbon footprint

AI and Systems Engineering over time

Systems Engineering

Bell Telephone Labs / GM: Systemdenken, ***Term Systems Engineering (SE)***

Information Systems (IS): development processes

NATO Software Engineering Conferences

V-Modell → V-Modell 97 → V-Modell XT

Agile SW development and operation e.g., SCRUM, DevOps, MLOps

SE in literature

SE process models
SE in practice: u.a. Apollo

INCOSE SE Handbooks V1-V5

AI Systems Engineering → PAISE®

1940 1950 1960 1970 1980 1990 2000 2010 2020

ML, SVM, neural nets (applications), e.g. IBM DeepBlue

Deep Learning, u.a. ImageNet

ChatBots e.g. ChatGPT, Gemini

Turing Test: Can machines think ?

first AI Systemes: Logic Theorist (1956), ELIZA (1966), SHRDLU (1970)

Agentic AI → Internet of AI Agents

AI Systems, AI assistants + LLM: e.g. GPT-3/GPT-4, Gemini, OpenGPT-X

Big Data. Cloud Computing → data driven AI

Neural networks (math. model)

Expert systems, e.g. DEC R1

Dartmouth Conference: ***Term AI***

Semantic Web

Artificial Intellgence (AI)

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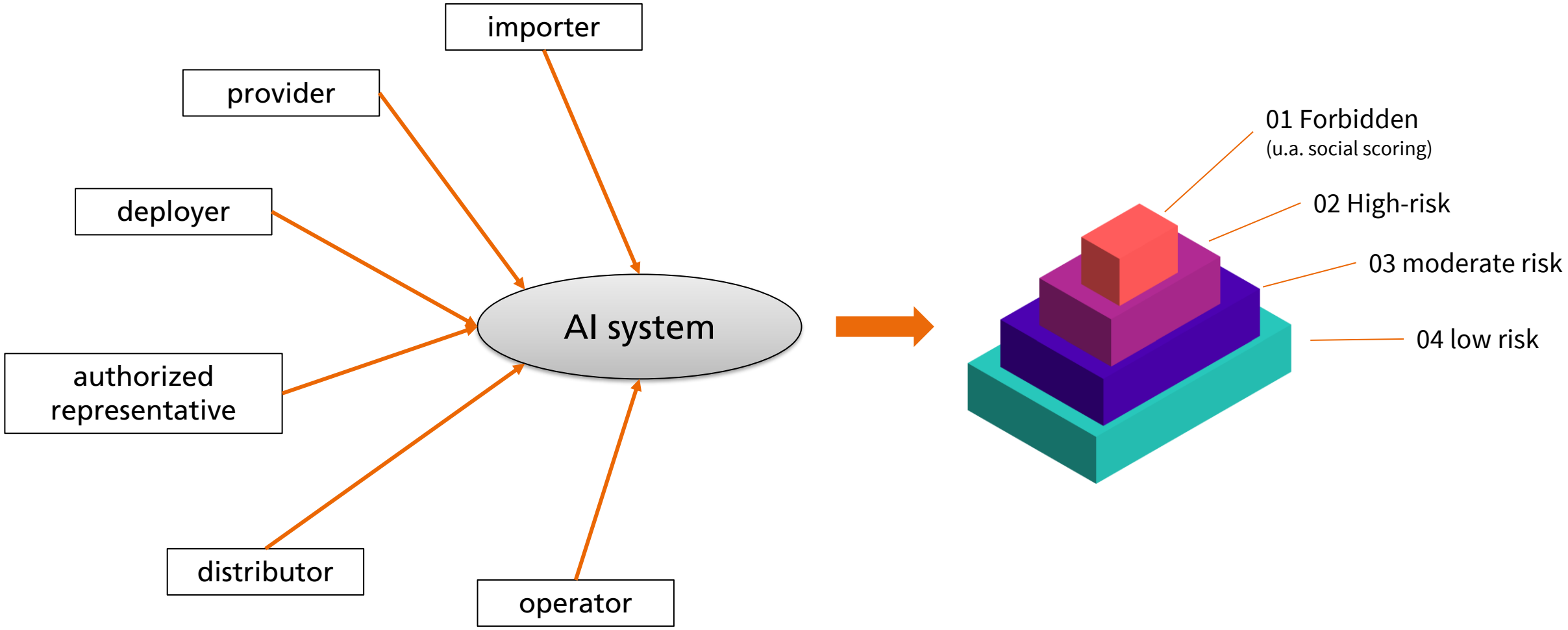
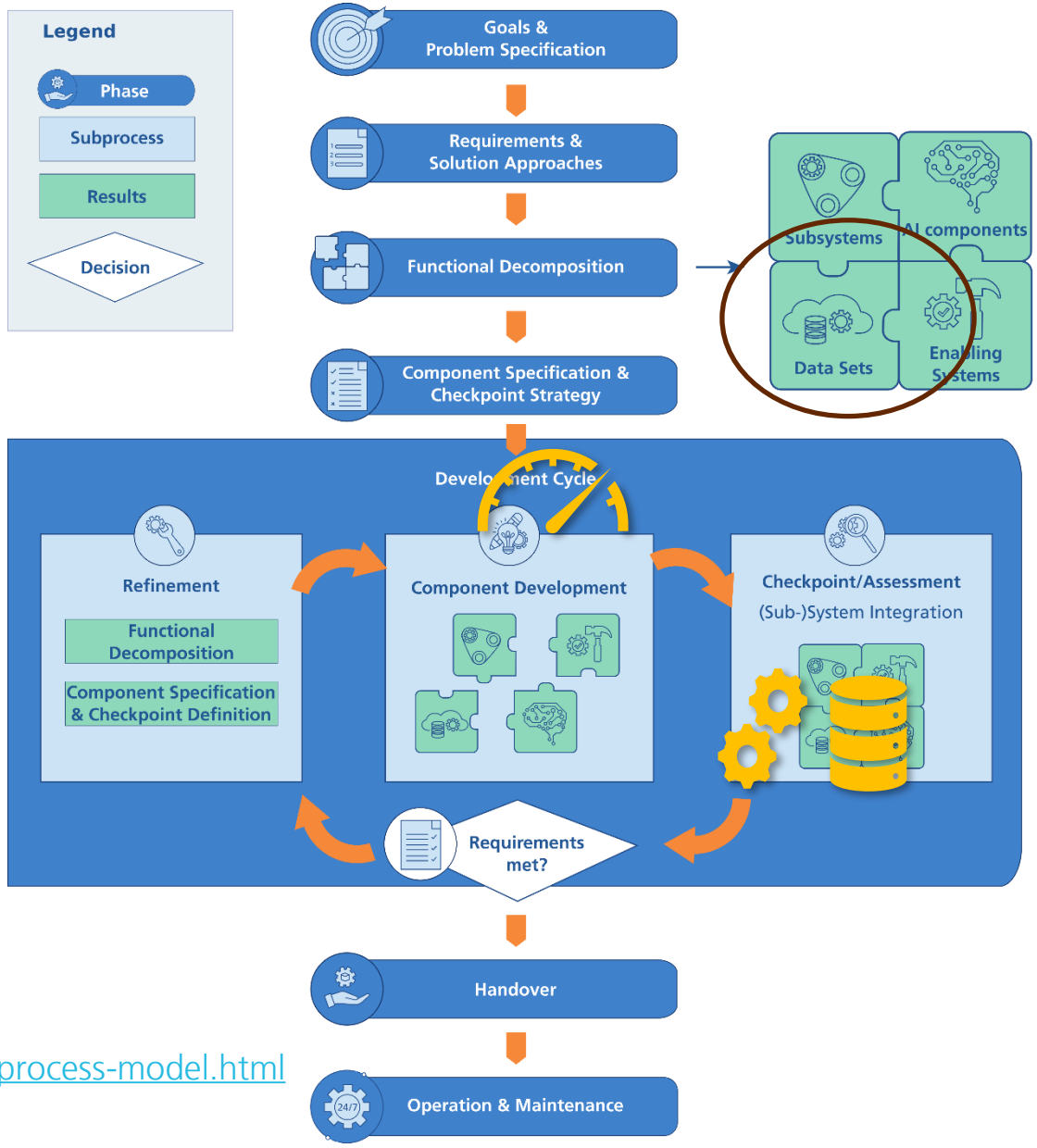
European AI Act: Roles and Risk Classes of AI Systems



PAISE®

Process Model for AI Systems Engineering

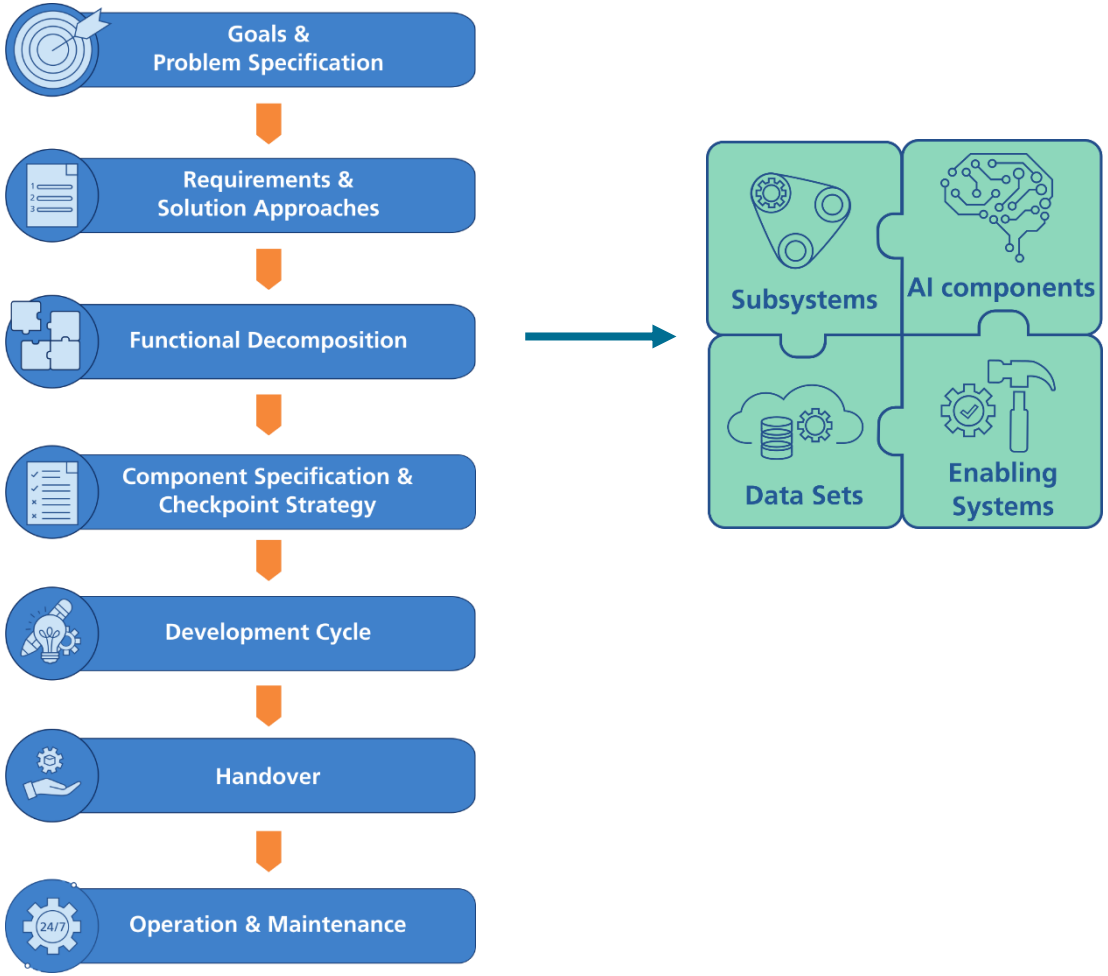


Datasets as sub-systems !

Information and Dokument:

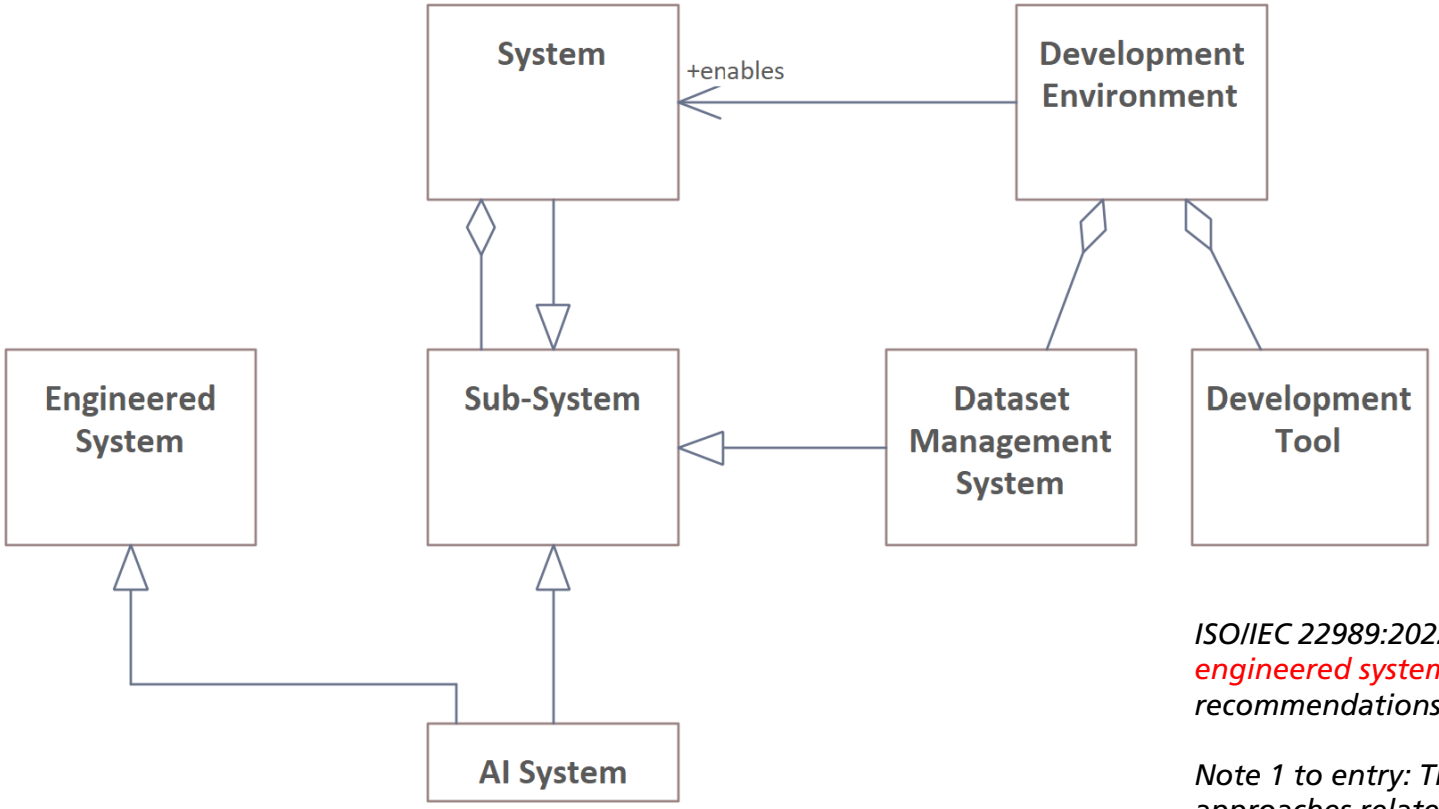
https://www.ki-engineering.eu/en/know-how-tools/paise-process-model.html

Basic Concept of PAISE®



- Sequential process model on the top level
  - E.g. waterfall model
  - Applicable to custom process models
- Decomposing the system
  - Hierarchical system architecture
  - Mixture of AI as well as traditional components
  - Each component has well defined functionality and interfaces
- AI sub system
  - Sub system utilizing AI techniques
- Data set
  - Data set utilized for development

Conceptual AI System Model (1)

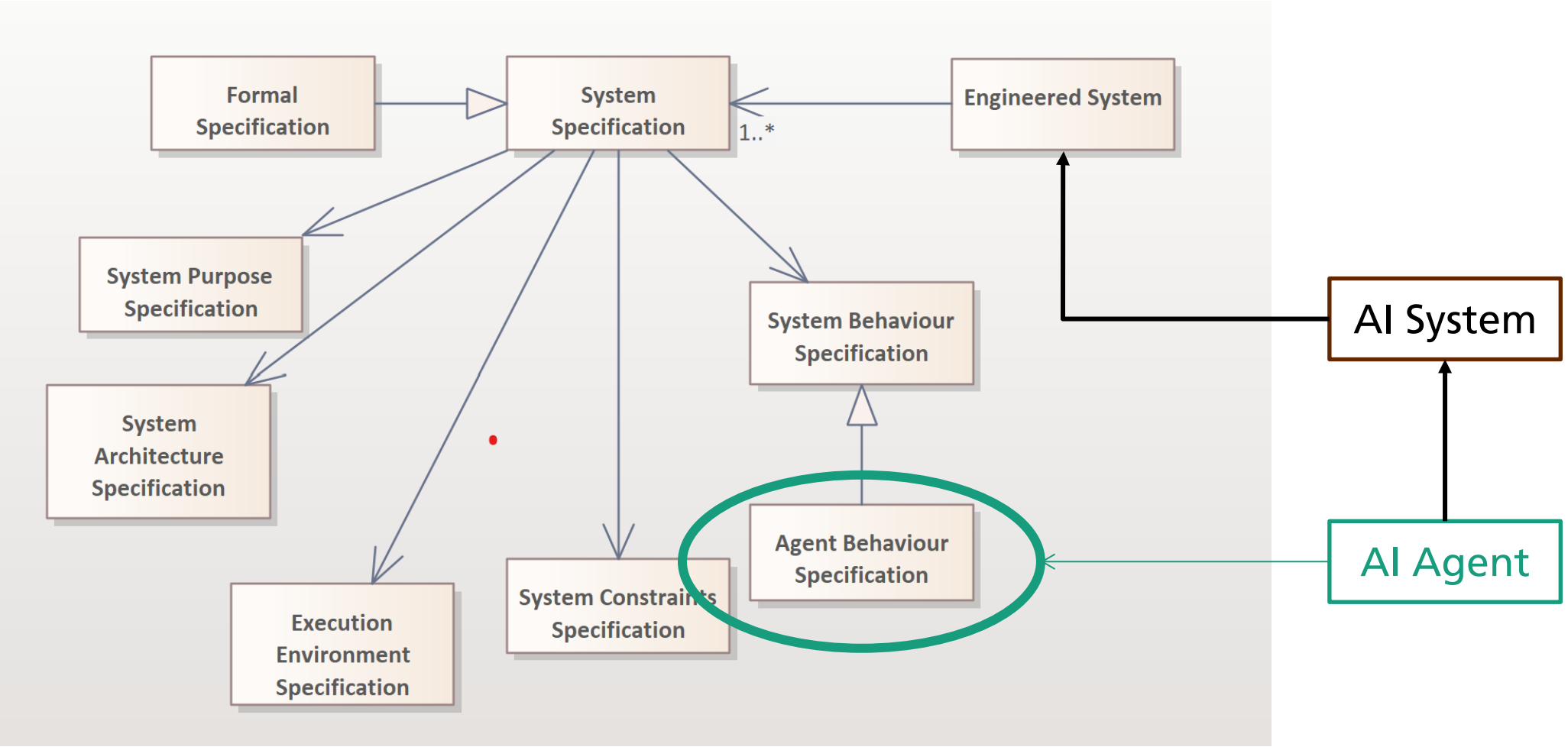


INCOSE: An **engineered system** *is a system designed or adapted to interact with an anticipated operational environment to achieve one or more intended purposes while complying with applicable constraints.*

ISO/IEC 22989:2022 : AI system =
engineered system that generates outputs such as content, forecasts, recommendations or decisions for a given set of human-defined objectives

Note 1 to entry: The engineered system can use various techniques and approaches related to artificial intelligence (3.1.3) to develop a model (3.1.23) to represent data, knowledge (3.1.21), processes, etc. which can be used to conduct tasks (3.1.35).
Note 2 to entry: AI systems are designed to operate with varying levels of automation (3.1.7).

Conceptual AI System Model (2)



Agent behaviour: An AI Agent is a specific kind of AI System

- **Embedded in an environment**
  - web, software system, physical world and continuously interacts with it.
- **Perception and action**
  - perceives the environment via inputs (text, sensors, API calls, logs, etc.).
  - acts on the environment via outputs (text responses, API calls, robot movements, etc.).
- **Goal-directed**
  - tries to achieve explicit goals or optimize a performance measure (e.g., minimize cost, maximize user satisfaction).
- **Autonomous**
  - makes decisions on its own, using its models, rules, and data, without being told every single step.
- **Reactive and pro-active**
  - reacts to changes (new input, new state).
  - plans ahead to reach long-term goals.

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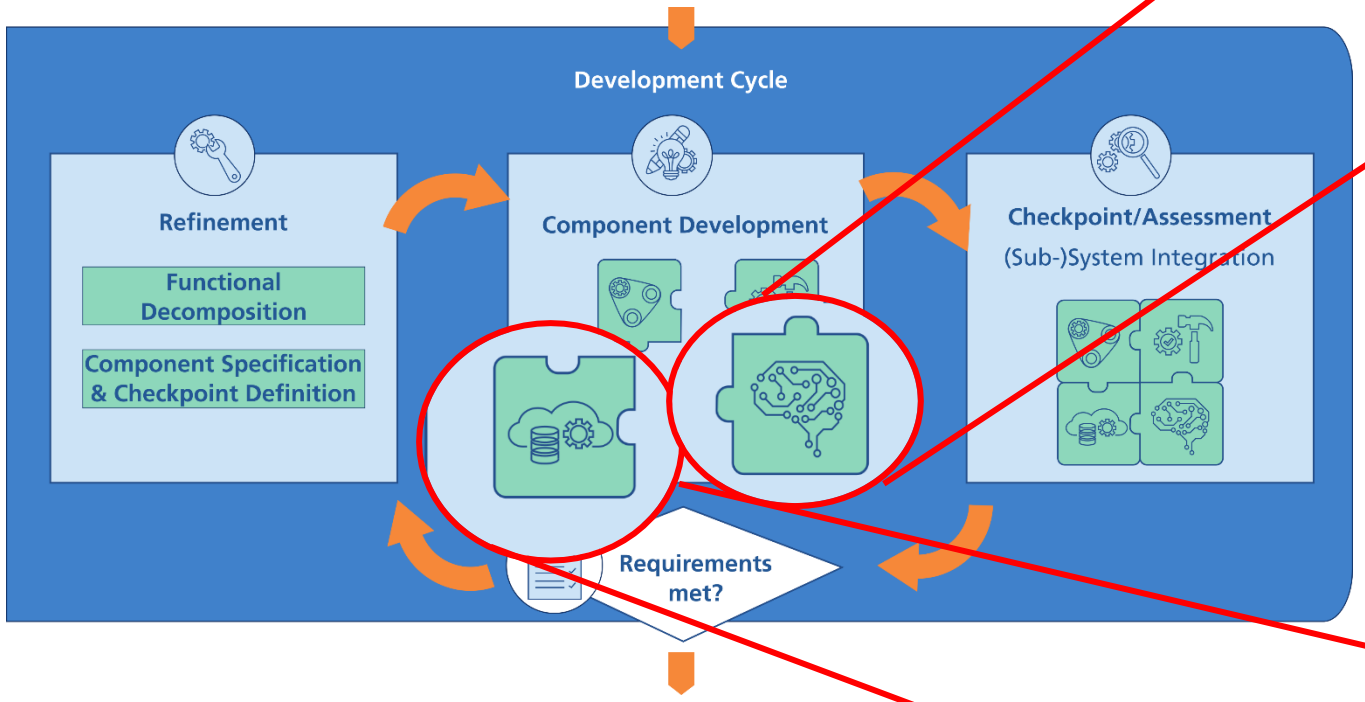
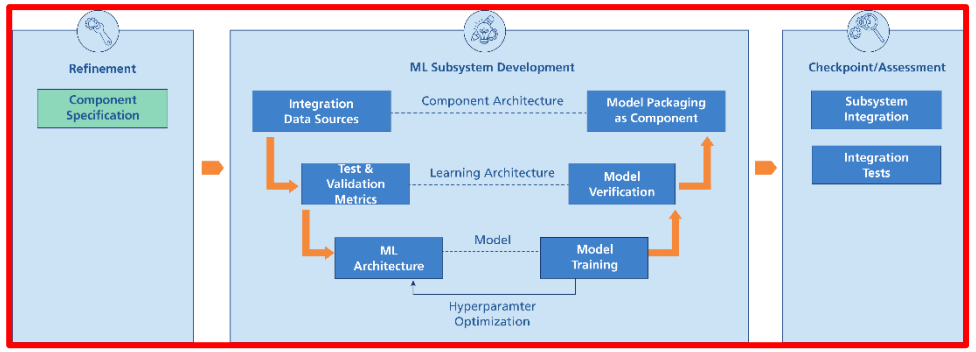
Component Development

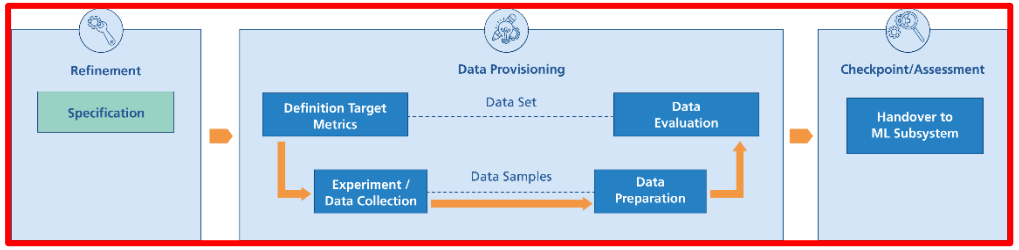




for AI systems

dedicated development cycles

for (training and test) datasets



Conclusive statements

- AI is just a technology and needs
  - methodology (→ AI Systems Engineering)
  - data (→ data spaces)

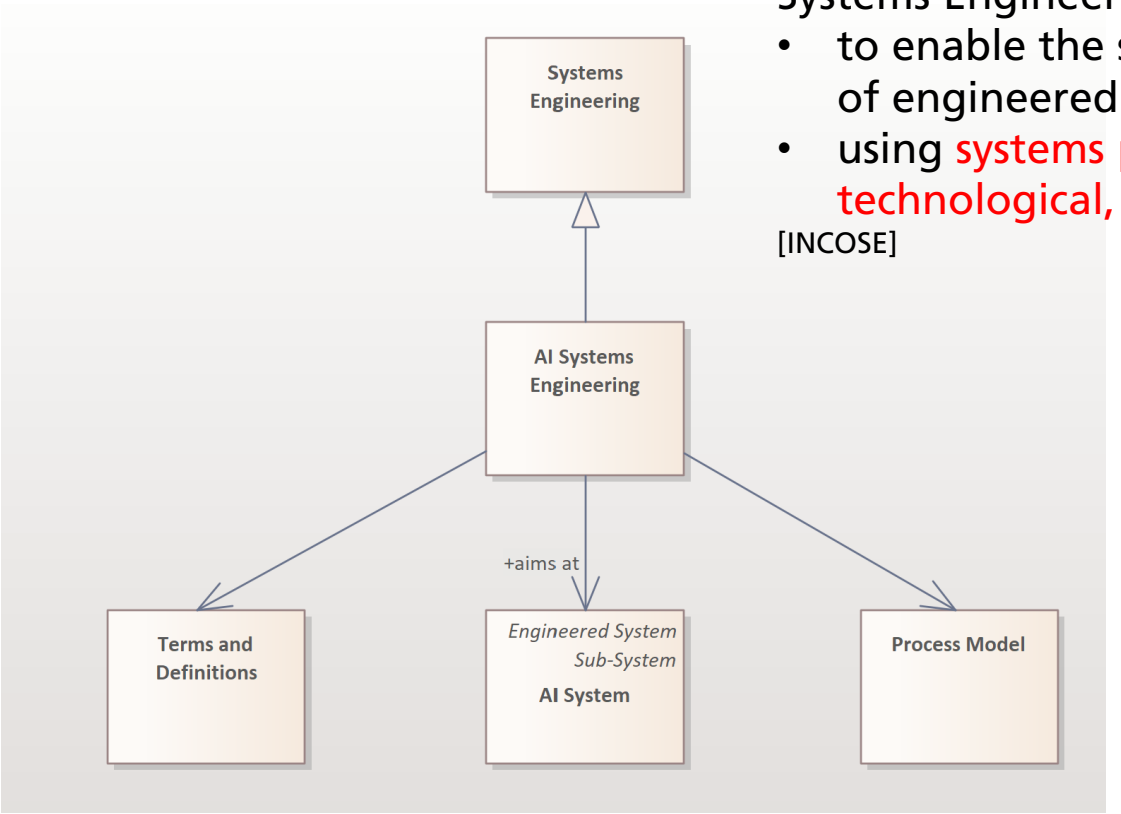
  to be systematically applicable in industrial applications
- Openness based upon standards is key for efficiency and flexibility
  - IEC 63278 Asset Administration Shell AAS
  - Data space protocol
- Compliance by design
- Data spaces needs to be extended towards sharing of AI Assets (data, AI models and computing resources)

AI Systems Engineering – a new engineering discipline ? (1)

An *engineering discipline* is a **distinct, organized branch of engineering** that focuses on a specific type of technical problem, methods, and applications, with its own:

- Body of knowledge
  - Theories
  - Models
  - Tools
- Methods and design practices
- Standards and regulations
- Typical application domains

AI Systems Engineering – a new engineering discipline ? (2)



Systems Engineering is a transdisciplinary and integrative approach

- to enable the successful realization, use, and retirement of engineered systems,
- using systems principles and concepts, and scientific, technological, and management methods

[INCOSE]

AI Systems Engineering addresses the

- systematical development and operation of
- AI systems (AI-based solutions as part of systems)
- that master complex tasks.

[CC-KING]

ISO/IEC 22989:2022
IT – AI – AI concepts and terminology

https://www.iso.org/standard/74296.html.

European AI Act.

PAISE®

Process Model for AI Systems Engineering.

Two sides of the same coin

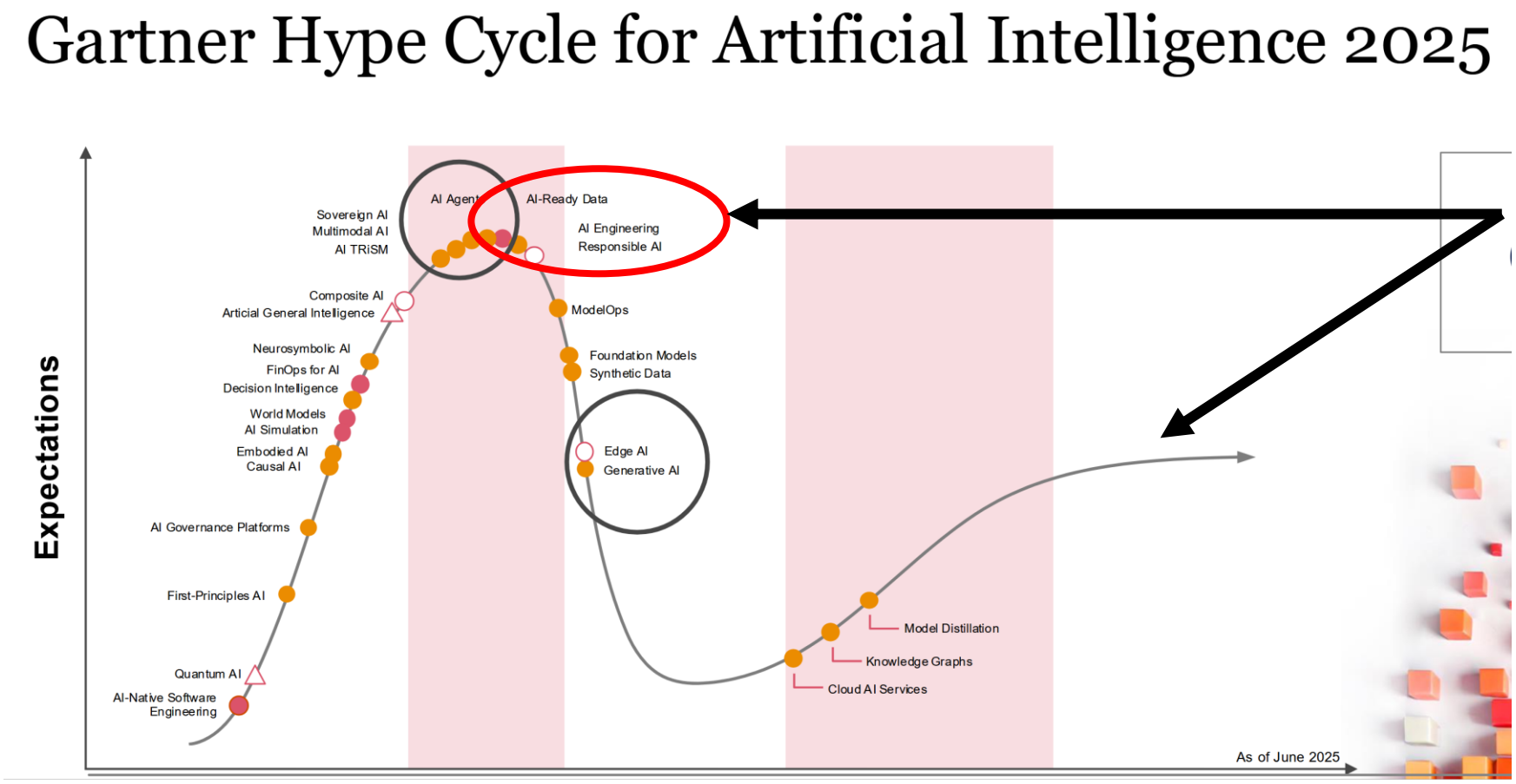
need for an engineering methodology

need for data

AI Systems Engineering

Data Spaces

Motivation global



AI Systems Engineering is key to survive the valley of death.

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CC-KING Competence Center KI-Engineering

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